Extending Place-Based Innovation Policy to Rural Areas: Using Oaxaca-Blinder Decomposition to Identify Endowment Deficiencies

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CES 26-48 **July 2026**

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Abstract

Place-based policy is premised on the idea that productive investments that address endowment deficiencies may release latent comparative advantages, stimulating growth while reducing regional disparities. Extending place-based policy to the domain of innovation is conceptually challenging given conventional wisdom that settlement size is the principal determinant of agglomeration economies that fuels innovation. However, growing concern that more places are being left behind by the innovation economy has prompted greater interest in ways to expand the geography of innovation. Using confidential firm-level data from the 2018 Annual Business Survey, this study examines differences in self-reported innovation activity between firms in rural and urban counties to identify possible endowment deficiencies. We do this by employing the Oaxaca-Blinder decomposition method to quantify the urban-rural innovation gap to better understand the relative contributions of firm characteristics, owner characteristics, and county-level factors.

Keyword: Place-based innovation policy, Rural–urban innovation gap, Oaxaca–Blinder decomposition, Annual Business Survey

JEL Classification: O31, R11

* Any views expressed are those of the authors and not those of the U.S. Census Bureau. The Census Bureau has reviewed this data product to ensure appropriate access, use, and disclosure avoidance protection of the confidential source data used to produce this product. This research was performed at a Federal Statistical Research Data Center under FSRDC Project Number 2512. (CBDRB-FY23-P2512-R10889, CBDRB-FY25-P2512-R12163, and CBDRB-FY25-P2512-R12626)

1. Introduction

The rural-urban divide in innovation and economic development is a pressing issue in the United States (Partridge 2020; Conroy, Deller, and Callon 2022). Metropolitan areas significantly outpace rural areas across major innovation indicators like R&D spending, patenting, and high-tech firm formation (Freshwater 2012). This gap poses challenges for economic growth and competitiveness of rural regions. Addressing innovation disparities between urban and rural areas has become a policy priority for fostering inclusive economic growth (Dabson 2011).

Compared with voluminous studies on innovation mostly in metropolitan areas, prior literature on the rural-urban innovation gap using quantitative methods is still scarce. Using patent applications as a measure of innovation output, quantitative studies consistently show lower innovation outputs in rural areas at the regional level (Aryal et al. 2018; Dotzel and Wojan 2022; Wojan, Dotzel, and Low 2015; Wojan and Parker 2017). However, patent applications do not provide a sufficient statistic to account for all innovation activities, including research and development (R&D) spending and personnel, new products and services, new processes and methods in logistics, management, and marketing (Acs and Audretsch 1993; Acs, Anselin, and Varga 2002). In addition, the common use of population to normalize patenting rates results in a bias that favors urban areas and disadvantages rural areas that tend to have higher dependency ratios and also have employment concentrated in sectors that do little or no patenting (Dotzel and Wojan 2022; Wojan 2022a). Moreover, rural areas may not be lagging urban areas in all aspects of innovation. For example, Goetz and Han (2020) construct a “latent innovation index” that quantifies hidden or otherwise unmeasured market and process innovations that occur within counties. With this new index, they find that many nonmetro areas exhibit comparatively high

levels of innovation potentials.[1] More generally, self-reported positive innovation indicators such as those in the EU's Community Innovation Survey and the Annual Business Survey in the US also suggest smaller differences in urban-rural innovation disparities than commonly assumed (Fitjar and Rodríguez-Pose 2020; Wojan 2022b).

This study addresses the gap in the literature through an exploratory analysis of rural-urban differences using firm-level self-reported innovation activities using the 2018 Annual Business Survey (ABS) that provides comprehensive information on nonfarm businesses. Our primary research method is the Oaxaca-Blinder (OB) decomposition that quantifies the innovation gap between rural and urban counties in the United States and examines the relative contributions of firm characteristics, firm owners' characteristics, and local contextual factors. Our work makes three key contributions. First, we provide new evidence on rural-urban innovation disparities using recent, nationally representative survey data. Second, our firm-level analysis offers additional insights beyond regional studies (e.g., county-level studies). Third, the decomposition approach lets us explore potential drivers of the innovation gap. Findings can inform policies to support innovation and economic development in rural areas.

In this paper, we define rural counties based on the 2013 Rural-Urban Continuum Code (RUCC) of Economic Research Service, U.S. Department of Agriculture (USDA ERS). Urban counties are RUCC = 1–3, and rural counties are RUCC=4–9. The paper is structured as follows. We first review prior literature on rural-urban innovation differences. Next, we describe our data and methods. Results are then presented, followed by a discussion of implications for research and practice. The conclusion summarizes key findings and contributions.

[1] Throughout the paper, urban or metro refers to areas with Rural-Urban Continuum Codes (RUCC) 1-3, while rural or nonmetro refers to areas with RUCC 4-9.

2. Literature Review

The existence of a substantial innovation gap between rural and urban regions is well established in prior research. Studies using regional-level data consistently find that urban areas significantly outperform rural areas on quantitative innovation indicators such as R&D investments, patenting rates, high-tech firm formation, and venture capital funding (Malecki 1997; Barkley, Henry, and Li 2004). Recent studies demonstrate composition effects explain a significant portion of lower rural patenting rates, due to smaller shares of engineers and patent-intensive industries. However, inherent urban advantages within detailed industries still persist. (Wojan, Dotzel, and Low 2015).

In terms of innovation types, urban regions are assumed to focus more on developing radical new technologies, while rural innovation are assumed to concentrate on adopting and incrementally improving existing products and processes. Using qualitative methods, Deakins and Bensemann (2019) find that small firms in rural environments can be just as innovative as their counterparts in urban environments, and rural firms tend to pursue incremental innovations due to resource constraints and risk aversion. Recent firm-level studies using government survey data find only a modestly lower self-reported innovation rate among rural firms (Wojan and Parker 2017; Wojan 2022).

Several interconnected factors help explain the rural-urban gap. Urban advantages include knowledge spillovers from dense proximity (Audretsch and Feldman 1996), specialized labor pools, and access to amenities and transportation infrastructure. Rural disadvantages include lack of critical mass, distance from networks and resources, and lagging innovation infrastructure (Acs and Malecki 2008; Goetz and Han 2020; Liu 2022). There is also evidence of spatial sorting of innovative firms (Li et al. 2016). Young startups focused on innovation disproportionately locate in diverse urban cores to benefit from knowledge spillovers. More

standardized, routine firms tend to spread to suburban or peripheral areas over time to reduce costs as they mature (Duranton and Puga 2000; Rosenthal and Strange 2010). This sorting aligns with theories about the drivers of firm location choice.

Closely related to our study, Aryal et al. (2021) provided firm-level analysis of the rural-urban innovation divide using the 2014 Rural Establishment Innovation Survey (REIS). They found rural firms reported significantly lower innovation creation in terms of patent applications. However, the REIS data has a relatively small sample size of which firms in nonmetropolitan areas are oversampled (Wojan and Parker 2017, Wojan 2022a), casting their representativeness into question. Also, as discussed above, using patent applications as a measure of innovation is questionable. In contrast, Wojan (2022b) examines the self-reported innovation using the 2018 ABS data, mainly focusing on new-to-market innovation, referring to as new products or services that no other companies have provided to the market. Our study contributes to this emerging firm-level analysis of rural-urban innovation disparities also using the 2018 ABS data but investigating process and marketing innovation in addition to new-to-market innovation to assess where the urban-rural innovation gaps are greatest.

3. Data and Method

3.1 Innovation Variables from the 2018 Annual Business Survey

The primary data source is the 2018 Annual Business Survey, with the data reference year 2017, conducted jointly by the U.S. Census Bureau and the National Center for Science and Engineering Statistics (NCSES) within the National Science Foundation (National Center for Science and Engineering Statistics 2020). The ABS includes all nonfarm employer businesses filing the 941, 944, or 1120 tax forms, accounting for approximately 850,000 firms. The survey

provides comprehensive information on economic and demographic characteristics for businesses and business owners. The 2018 ABS has a special module for innovation, covering a wide range of topics, such as the types of innovation, factors impeding innovation, and reasons for not innovation, among others. The survey also includes a research and development (R&D) module exclusively for microbusinesses with one to nine employees.

Pertinent to this study, we focus on the survey questions about self-reported types of innovation, namely product innovation, process innovation, and organizational and marketing innovation. (See the definition of these innovation types on page 17 of the 2018 ABS survey questionnaire in supplemental materials.) Process innovation encompasses any new methods of manufacturing, logistics, distribution, and supporting activities. Organizational and marketing innovation include new business practices, such as supply chain, knowledge, and quality management, new design and packaging, and new product promotion, placement, and pricing methods, etc. We create the dummy variables indicating new-to-market, process, and organizational and marketing (hereafter, we simply refer it to as marketing) innovation, which are the dependent variables in the regression and decomposition analysis.

We restricted our sample from the ABS to include only single-unit firms in order to mitigate potential issues arising from the locations of multi-unit firms (Wojan 2022). This decision was driven by the unique challenges posed by multi-unit firms, where the geographical characteristics of the reporting unit may differ substantially from other units of the firm. This makes it difficult to accurately attribute local contextual factors to the firm as a whole. ABS includes an establishment file with the location of the constituent units of a firm, but the innovation practices in any particular unit is unknown. Thus, we are unable to assign innovation variables to specific establishments within multi-unit firms.

3.2 Data for Explanatory Variables

We consider three main factors when selecting explanatory variables: characteristics of business owners, characteristics of businesses, and local contextual factors. We select variables that can effectively explain personal motivations and conditions for innovation, practices in businesses that require innovation, and location factors that attract innovative individuals and businesses and create an environment conducive to innovation. The rationale of the Oaxaca-Blinder decomposition analysis is to identify innovation gaps associated with difference in endowments that are observed in the data, often referred to as explained or observed factors, and innovation gaps associated with differences in the returns to these endowments across urban and rural areas, often referred to as unexplained or unobserved factors.

We extract explanatory variables that represent business owners' characteristics and business characteristics from the 2018 ABS. The demographic variables include owners' age (younger than 45, older than 65, or in between), gender (female or male), ethnicity (Hispanic, non-Hispanic white, or other non-white non-Hispanic ethnic groups), foreign-born status, and education (with a bachelor's degree or not and with a degree in Science, Technology, Engineering and Mathematics (STEM) majors or not). We also have a variable indicating if a firm has multiple owners.[2] Business characteristics include variables for employment size (1–9, 10–250, or more than 250 employees), a manufacturing sector indicator, whether the firm reported in-house R&D activities, and family businesses jointly owned by family members. In

[2] The ABS provides firm-owner characteristics for the four largest share firm owners. We create binary variables (dummies) to indicate the presence of specific characteristics among these firm owners. For example, the "female" dummy is set to one if at least one of the firm owners is female. Similarly, the "STEM degree" dummy is assigned a value of one if any of the firm owners holds a STEM degree, and the "foreign-born" dummy is one if any of the owners were born outside the United States.

addition, we include a variable indicating if a firm has adopted any cloud services, such as data storage and management, servers, data analytics, etc. (Han, Wojan, and Goetz 2023)

The final set of explanatory variables focuses on local contextual factors that influence the spatial distribution of innovative firms. As the rural-urban divide is also a location factor, we choose both positive and negative location variables for rural innovation. We include local latent innovation index proposed by Goetz and Han (2020). This index measures innovation potentials in a region, using the prevalence of existing industries and expected local interaction based on the direct requirements coefficients from the national input-output table. Using U.S. county-level data, the authors find substantial latent innovation capacities exist across rural America that are overlooked by conventional metrics, such as patent applications. The second location factor related to rural innovation is inventive class employment proposed in the papers of Wojan, Dotzel, and Low (2015) and Dotzel and Wojan (2022). Taking an occupational perspective, this variable measures the potential population that are mostly likely to participate in a patent application based on the association between 59 detailed occupations and patenting (Dotzel and Wojan 2022). The authors argue that this variable should be the denominator in a measure proxying patenting productivity rather than the nearly universal use of patents per capita. Using this variable, the authors find that rural counties have far fewer inventive occupations per capita, explaining more than half of the rural-urban patenting gap even after controlling for other factors. Two human capital variables in the analysis include the share of those over 25 in county that majored in a science and engineering discipline from the American Community Survey, and the percentage of college students in the county majoring in STEM (Chetty et al. 2017). The latent innovation, inventive class, science and engineering graduates, and STEM major variables

control for and assess whether a potentially more innovative environment stimulates firm-level innovation.

The third important local factor is social and creative human capital that are both dependent on interaction of residents in a locale. The social capital index developed by Rupasingha, Goetz, and Freshwater (2006) demonstrates substantial geographic variation, with the highest production of social capital in rural counties and smaller communities. This has implications for rural-urban differences in innovation. Social capital may provide connections and trust that facilitate idea sharing and risk-taking necessary for innovation. The greater social capital production observed in rural areas suggests rural communities have strong bonding social capital within their local networks. Third spaces such as coffee shops, bars, or restaurants are believed to be important venues for the creation of bridging social capital. The walkability of a locale may also affect serendipitous interaction indicated by the share of workers commuting by foot or bike. Endowments of creative human capital are measured using an arts avocation index along with the share of the population over 25 that majored in the arts (Wojan 2025). We also include the broadband adoption rate from Federal Communication Commission, the county typology for retirement destination and natural amenity scales from the USDA Economic Research Service, as well as median home value from the 2013-2017 five-year American Community Survey. Finally, a policy variable of interest is the amount of Small Business Innovation Research (SBIR) awards per 10,000 population.

3.3 The Oaxaca-Blinder decomposition method

Oaxaca-Blinder decomposition allows assessing the relative contribution of endowment level and the returns to endowments to local innovation. The baseline regression model is a logistic model $\Pr(Y_{ic} = 1) = \text{logit}(X_{ic}\beta_c + \varepsilon_{ic})$, with the subscripts for firm i and county type

c={urban, rural}. To perform the Oaxaca-Blinder (OB) decomposition, we estimate the model separately for firms in urban and rural counties. The OB technique divides differences-in-mean in the dependent variable between firms in urban and rural counties into two components: the "explained" part accounts for observable characteristics included in the model, and the "unexplained" part attributes the difference-in-mean to any unobserved factors pertinent to group membership (Oaxaca 1973; Blinder 1973).

The decomposition model can be expressed as:

$$\underbrace{\overline{Y}_U - \overline{Y}_R}_{\text{Difference in mean}} = \underbrace{\sum_{k=1}^{K} (\overline{X}_{U,k} - \overline{X}_{R,k})\hat{\beta}_{*,k}}_{Explained} + \underbrace{\sum_{k=0}^{K} [\overline{X}_{U,k}(\hat{\beta}_{U,k} - \hat{\beta}_{*,k}) + \overline{X}_{R,k}(\hat{\beta}_{*,k} - \hat{\beta}_{R,k})]}_{Unexplained} \quad (1)$$

where $\overline{Y}_c$, c={U, R} is the mean innovation outcome (i.e., the proportion of firms with a certain type of innovation). The first term on the right-hand side captures the part of the rural-urban innovation gap attributable to differences in observable variables, which is also called the "endowment effect. The second term reflects the unexplained gap, also called the "coefficient effect", which may represent structural barriers or advantages associated with rural location. $\hat{\beta}_{*,k}$ is the coefficient for a reference case, for which we use the coefficient estimated from a pooled regression by stacking all rural and urban firms as suggested by Jann (2008).

This OB decomposition provides a useful descriptive framework for this study's goal of documenting rural-urban innovation gaps and understanding their underlying sources when controlling for key firm differences. From equation (1), we can attribute the explained and unexplained parts to each variable's contribution, i.e., $(\overline{X}_{U,k} - \overline{X}_{R,k})\hat{\beta}_{*,k}$, and $\overline{X}_{U,k}(\hat{\beta}_{U,k} - \hat{\beta}_{*,k}) + \overline{X}_{R,k}(\hat{\beta}_{*,k} - \hat{\beta}_{R,k})$. The interpretation of the explained part in the Oaxaca-Blinder decomposition is as follows: if rural firms have the same average value in X_k as urban firms, the innovation differences attributed to X_k would diminish. On the other hand, in the unexplained

part, even under the same counterfactual assumption that rural and urban firms have the same average value in X_k, the urban-rural difference remains due to the discrepancy in the coefficient of the variable. However, we cannot deduce causal effects from the OB method; instead, it is a valuable first step in a rigorous empirical analysis.

4. Results

Table 1 compares variable means (and standard deviations) between firms in metro and nonmetro counties in our sample. Regarding innovation activities, firms in urban counties exhibit significantly higher means for all three types, although the difference is smallest for new-to-market innovation. In terms of business owners' characteristics, firms in rural counties primarily surpass those in urban counties in three out of nine aspects: female owners, owners older than 65, and multiple owners. For firms' characteristics, firms in rural counties are less likely to adopt cloud computing but more likely to be small (nine or fewer employees), engaged in manufacturing, and jointly owned by family members. As for county characteristics, with the exception of being located in retirement-destination counties, most aspects exhibit substantial differences. Rural counties have a higher social capital index but fall behind in broadband adoption rate, inventive class employment, latent innovation index, and median home value. Interestingly, urban counties have a higher natural amenity index, potentially due to rapid growth in the urban sunbelt.

[Table 1 around here]

Figure 1 depicts the estimated coefficients for urban and rural counties in regression estimation. The coefficients represent factors influencing innovation activities. Most of the factors we considered are positive for innovation. Notably, in-house R&D, adopting cloud computing, being manufacturing firms, having large firm sizes, and having young business

owners are statistically significant factors for all types of innovation activities. Conversely, there are negative factors, including old business owners and multiple owners in rural firms. Furthermore, firms with female owners, both urban and rural, exhibit relatively lower levels of process innovation activities compared to male owners. For detailed estimated coefficients, please refer to Tables A1 and A2 in the appendix.

[Figure 1 around here]

Figures 2–4 present the outcomes of the Oaxaca-Blinder decomposition for three types of innovation: new-to-market, process, and marketing. In each figure, panel (a) illustrates the total difference-in-mean for the dependent variable, along with the explained and unexplained portions of the total difference between firms in urban and rural counties. Panels (b) and (c) further break down the explained and unexplained difference by each explanatory variable. The order of explanatory variables is determined by the magnitude of the explained decomposition. Notably, as observed in panel (a) across all figures, the model effectively accounts for the variations in innovation between urban and rural counties. For the first two types of innovation, the explained components surpass the total differences. Regarding marketing innovation, the model explains 87% of the total differences.

Regarding the contribution of each explanatory variable, it is noteworthy that adopting cloud computing stands out as the most prominent factor for two types of innovation. This is supported by the substantial positive difference-in-mean for this factor, as shown in Table 1, and its estimated coefficients, as depicted in Figure 1.[3] Along with the broadband adoption rate, these

[3] It is possible that cloud adoption is merely a reliable indicator of the most innovative firms, suggesting that cloud adoption by itself may not increase innovativeness of a business. Han, Wojan, and Goetz (2023) control for innovation orientation using propensity score match and endogenous treatment models and still identify an innovation-enabling effect of cloud computing services adoption.

effects may be amenable to investments to reduce the urban-rural digital divide. In-house R&D is another prominent factor associated with the urban-rural innovation gap that may be highly endogenous. The county-level latent innovation index plays a crucial role in new-to-market product innovation and process innovation but is only marginally significant in the marketing innovation decomposition. Since this variable is dependent on the complexity of local input buying relationships, it may be less amenable to policy. Inventive class employment seemingly reduces the urban-rural innovation gap in both new-to-product and marketing innovation. For product innovation this appears to result from minimizing the negative association between the inventive class and self-reported innovation—where thick markets reduce the probability of reporting a novel innovation. In contrast, the inventive class is positively associated with marketing innovation in rural areas but not in urban areas, reducing the urban-rural innovation gap due to one of the few statistically significant unexplained factors. Business owners' characteristics demonstrate varying contributions to different types of innovation. For instance, foreign-born ownership positively and significantly contributes to the new-to-market and process urban-rural innovation gap but has less impact on marketing innovation. Significant ameliorating factors, reducing urban-rural explained difference in the dependent variable include that aforementioned inventive class for product innovation, manufacturing and family jointly owned for all three types of innovation; and social capital for marketing innovation. Lastly, only a few factors make a significant contribution to the unexplained difference in the dependent variable, such as foreign-born owners for process innovation and inventive class employment for marketing innovation.

[Figures 2-4 around here]

5. Conclusion

This study provides new evidence on disparities in innovation activities between rural and urban firms in the United States. Using nationally representative data from the 2018 Annual Business Survey, we find that firms in urban counties exhibit significantly higher means across three innovation outcome measures: new-to-market, process, and marketing innovation. To gain insights into the drivers behind these rural-urban gaps, we employ the Oaxaca-Blinder decomposition technique. This allows us to quantify how much of the divide is explained by differences in firm characteristics versus unexplained factors related to location. Our results reveal the model accounts for most of the observed innovation gaps.

While our descriptive analysis provides valuable insights, further research could apply causal inference techniques to model the complex relationships between geography, firm characteristics, and innovation outcomes. As rural regions continue to reinvent their economic roles amidst global change, harnessing innovation will be critical for prosperity. Understanding its distinct nature in rural areas is essential for effective policies. This study contributes firm-level evidence to inform strategies for inclusive, innovation-led development.

Our results support the hypothesis that a considerable share of the urban-rural innovation gap is due to rural endowment deficiencies in infrastructure and human capital. The fact that the urban-rural innovation gap is overwhelmingly associated with endowment deficiencies rather than an urban advantage with respect to higher innovation returns to those endowments does suggest a potential role for place-based innovation policy. Novel findings regarding rural endowments that mitigate the innovation gap for some types of innovation such as the rural concentration of manufacturing, family-owned businesses, and social capital, suggest a number of possible routes for increasing the incidence of rural innovation.

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Funding Information: This work was supported in part by the United States Department of Agriculture (USDA) National Institute of Food and Agriculture (NIFA) under project #2023-51150-41113 and by the Pennsylvania State University and NIFA Multistate/Regional Research Appropriations under project #NE2249. This work is/was also supported by the USDA NIFA and Hatch Appropriations under Project #PEN05090 and Accession #7010745.

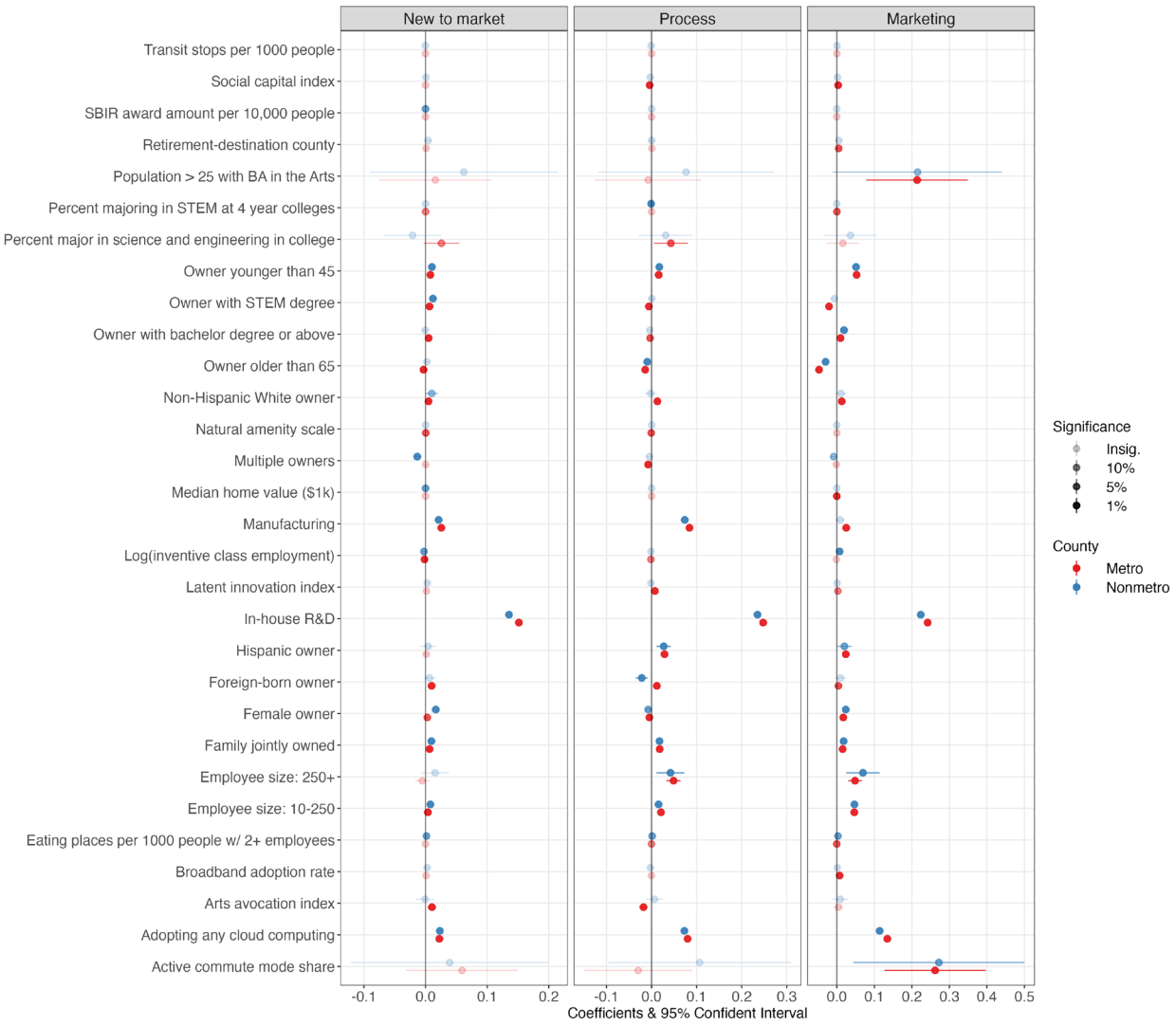
Tables and Figures

Table 1 Summary Stats by Rural-Urban, ABS sample weighted

Variable	Metro	Nonmetro
Dependent variables		
New-to-market innovation	0.090 (0.29)	0.080 (0.27)
Process innovation	0.166 (0.37)	0.145 (0.35)
Marketing innovation	0.227 (0.42)	0.193 (0.39)
Owners' characteristics		
Female owner	0.418 (0.49)	0.463 (0.50)
Hispanic owner	0.073 (0.26)	0.028 (0.16)
Non-Hispanic white owner	0.149 (0.36)	0.055 (0.23)
Foreign-born owner	0.208 (0.41)	0.063 (0.24)
Owner with bachelor's degree	0.541 (0.50)	0.417 (0.49)
Owner with STEM degree	0.230 (0.42)	0.186 (0.39)
Owner younger than 45	0.241 (0.43)	0.236 (0.42)
Owner older than 65	0.216 (0.41)	0.251 (0.43)
Multiple owners	0.405 (0.49)	0.456 (0.50)
Firms' characteristics		
Adopting any cloud computing	0.472 (0.50)	0.379 (0.49)
Employee size: 1-9	0.782 (0.41)	0.800 (0.40)
Employee size: 10-250	0.216 (0.41)	0.200 (0.40)
Employee size: > 250	0.002 (0.04)	0.001 (0.03)
Manufacturing	0.045 (0.21)	0.056 (0.23)
Family jointly owned	0.307 (0.46)	0.404 (0.49)
In-house R&D	0.096 (0.29)	0.071 (0.36)
County characteristics		
Retirement-destination county	0.151 (0.36)	0.132 (0.34)
Broadband adoption rate	4.108 (0.56)	2.850 (0.82)
Log(inventive class employment)	10.010 (1.48)	6.328 (1.12)
Latent innovation index	1.183 (0.65)	-0.062 (0.73)
Natural amenity scale	1.762 (3.47)	0.220 (2.60)
Social capital index	-0.588 (0.67)	0.156 (1.00)
Eating places with 100 people w/ 2+ employees	3.617 (31.69)	2.231 (2.058)
Active commute mode share	0.01676 (0.0167)	0.01697 (0.0158)
Transit stops per 1k pop	2.39 (13.84)	0.15 (1.168)
Arts avocation index	0.9907 (0.1621)	0.9586 (0.1678)
Percent majoring STEM at 4-year-college	2.932 (6.549)	2.748 (4.941)
Population >25 with BA in Arts	0.440 (0.046)	0.399 (0.055)
SBIR award amount per 10K population ($1m)	12.5 (1910)	0.52 (5.8)
Median home value ($1k)	281.800 (172.70)	148.200 (88.04)

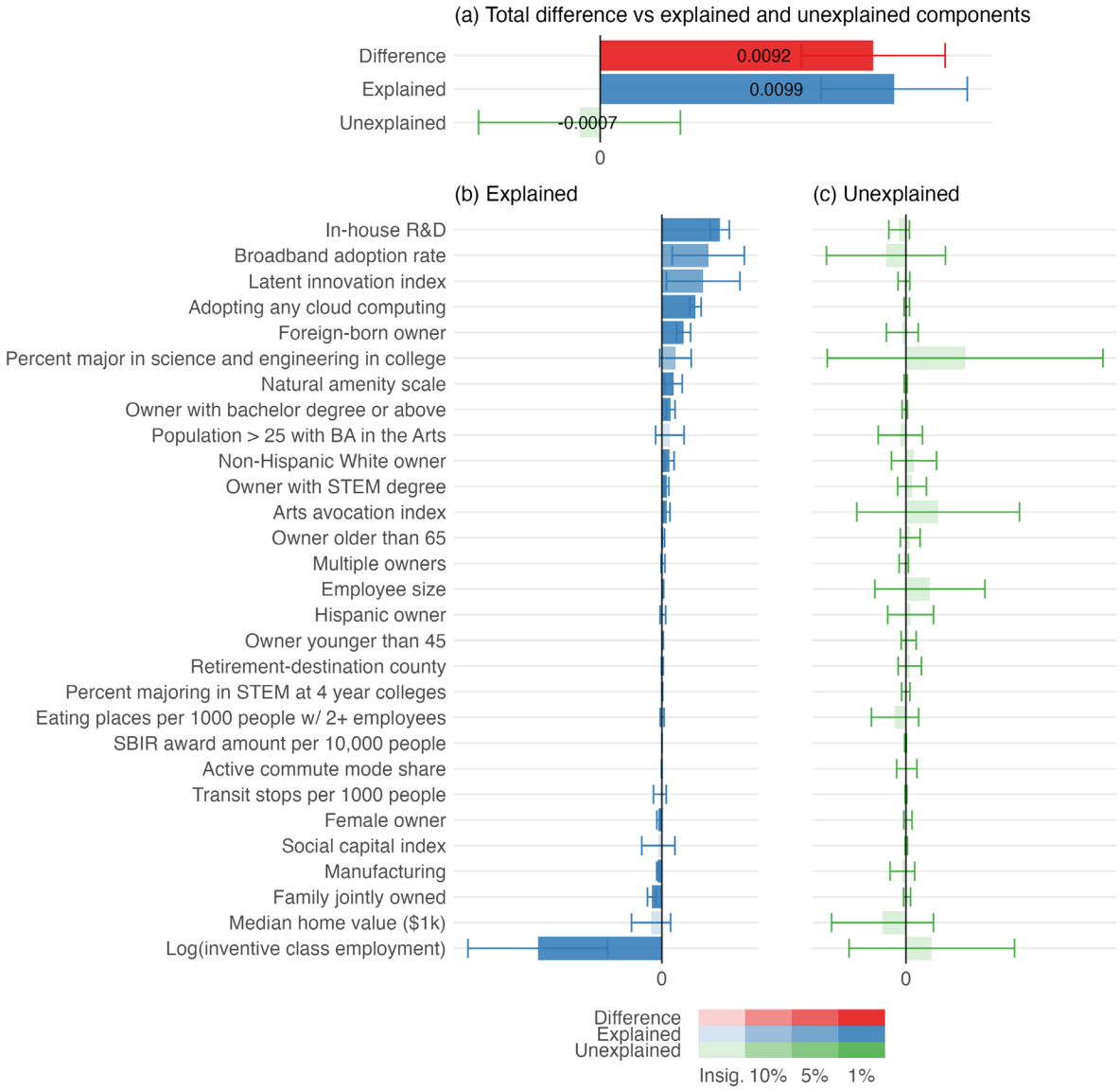
Notes: The number of firms in metro counties is 395,000, and that in nonmetro counties is 71,000. Columns 'Metro' and 'Nonmetro' show the mean and standard deviation (parenthesized) of each variable. All values are rounded according to the disclosure rules of the FSRDC. The ABS survey weights are used. The summary statistics of county characteristics may not accurately represent the true distribution at the county level because they are computed with the firm-level data.

Figure 1 Estimating Logit Models for Metro and Nonmetro Firms



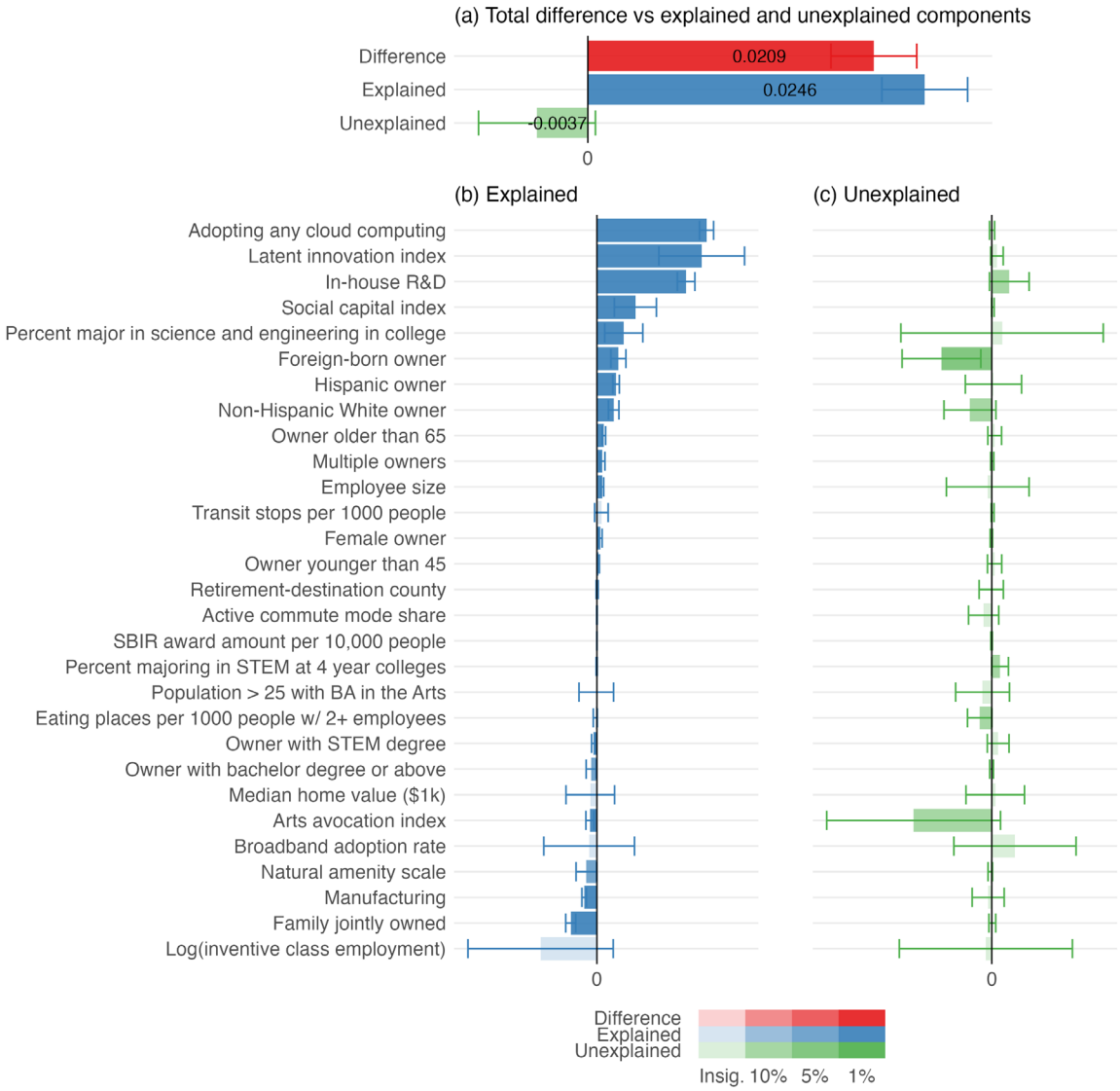
Notes: Estimated coefficients from separate logit models for metro and nonmetro firms for three innovation outcomes: new-to-market, process, and marketing innovation.

Figure 2 OB Decomposition for New-to-market Innovation



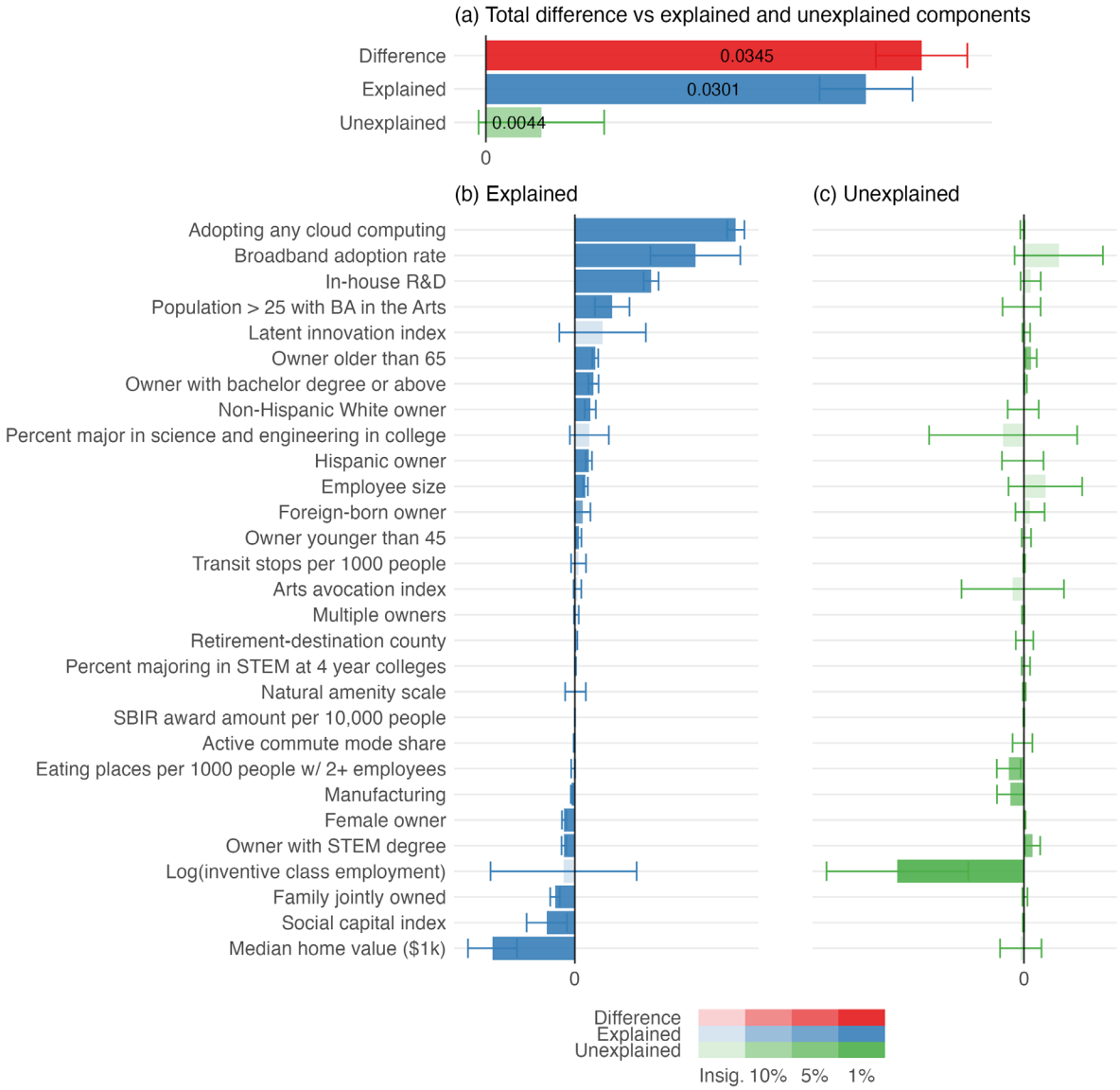
Notes: Oaxaca–Blinder decomposition of the rural–urban gap in new-to-market innovation, showing the total gap, explained and unexplained components, and the contribution of each explanatory variable.

Figure 3 OB Decomposition for Process Innovation



Notes: Oaxaca–Blinder decomposition of the rural–urban gap in process innovation, showing the total gap, explained and unexplained components, and the contribution of each explanatory variable.

Figure 4 OB Decomposition for Marketing Innovation



Notes: Oaxaca–Blinder decomposition of the rural–urban gap in marketing innovation, showing the total gap, explained and unexplained components, and the contribution of each explanatory variable.

Appendix Tables

Table A1 Estimated Coefficients for Nonmetro Firms

	New-to-market innovation	Process innovation	Marketing innovation
Female owner	0.01663***	-0.00746**	0.02382***
	(0.003)	(0.003)	(0.004)
Hispanic owner	0.0041	0.02701***	0.02031**
	(0.007)	(0.008)	(0.010)
Non-Hispanic White owner	0.01011*	-0.00201	0.011
	(0.005)	(0.007)	(0.008)
Foreign born owner	0.00648	-0.02166***	0.00993
	(0.005)	(0.007)	(0.007)
Owner with bachelor's degree or above	-0.00042	-0.00372	0.01923***
	(0.003)	(0.004)	(0.004)
Owner with STEM degree	0.01197***	0.00044	-0.0063
	(0.003)	(0.004)	(0.005)
Owner younger than 45	0.01041***	0.01728***	0.05134***
	(0.003)	(0.003)	(0.004)
Owner older than 65	0.00208	-0.00953***	-0.02998***
	(0.003)	(0.003)	(0.004)
Multiple owner	-0.01345***	-0.00437	-0.00849*
	(0.003)	(0.004)	(0.005)
Adopting any cloud computing	0.02322***	0.07282***	0.1141***
	(0.002)	(0.003)	(0.003)
Employee size: 10-250	0.00787***	0.01556***	0.04694***
	(0.003)	(0.003)	(0.004)
Employee size: > 250	0.01542	0.04204***	0.06970***
	(0.012)	(0.016)	(0.023)
Manufacturing	0.02126***	0.07372***	0.00908
	(0.003)	(0.005)	(0.006)
Family jointly owned	0.00949***	0.01762***	0.01812***
	(0.003)	(0.004)	(0.005)
In-house R&D	0.1355***	0.2351***	0.2237***
	(0.003)	(0.004)	(0.005)
Retirement-destination county	0.00409	0.00022	0.00525
	(0.003)	(0.004)	(0.005)
Broadband adoption rate	0.00246	-0.00244	0.00128
	(0.002)	(0.002)	(0.003)
Log(inventive class employment)	-0.00253**	-0.00124	0.00738***
	(0.001)	(0.002)	(0.002)
Latent innovation index	0.00269	-0.00122	0.00066
	(0.002)	(0.002)	(0.002)
Natural amenity scale	0.00021	0.00047	-0.00005
	(0.001)	(0.001)	(0.001)
Social capital index	0.00058	-0.00279	0.00194
	(0.001)	(0.002)	(0.002)
Median home value ($1k)	0.00004**	-0.00001	-0.00003
	(0.000)	(0.000)	(0.000)
Arts avocation index	-0.00067	0.00634	0.00886

	(0.008)	(0.010)	(0.011)
Active commute mode share	0.03917	0.1067	0.2721**
	(0.082)	(0.103)	(0.116)
Population >25 with BA in Arts	0.06215	0.07641	0.2152*
	(0.078)	(0.099)	(0.115)
Percent majoring science and engineering	-0.02106	0.03145	0.03635
	(0.024)	(0.030)	(0.036)
Percent majoring STEM at 4-year-college	0.00013	-0.00084***	-0.00008
	(0.000)	(0.000)	(0.000)
Transit stops per 1k pop	-0.00022	-0.00097	0.00043
	(0.001)	(0.001)	(0.001)
Eating places with 100 people w/ 2+ employees	0.00149**	0.00140**	0.00286**
	(0.001)	(0.001)	(0.001)
SBIR award amount per 10K pop.	2.27e-10**	1.05E-10	1.18E-10
	(0.000)	(0.000)	(0.000)

Notes: (1) The number of observations is 71000. (2) Significance levels: * 0.1, ** 0.05, *** 0.01. The ABS survey weights are used.

Table A2 Estimated Coefficients for Metro Firms

	New-to-market innovation	Process innovation	Marketing innovation
Female owner	0.00278**	-0.00483***	0.01723***
	(0.001)	(0.001)	(0.002)
Hispanic owner	0.00112	0.02879***	0.02395***
	(0.002)	(0.002)	(0.003)
Non-Hispanic White owner	0.00482***	0.01297***	0.01317***
	(0.002)	(0.002)	(0.002)
Foreign born owner	0.00983***	0.01186***	0.00427**
	(0.001)	(0.002)	(0.002)
Owner with bachelor's degree or above	0.00504***	-0.00320**	0.00986***
	(0.001)	(0.002)	(0.002)
Owner with STEM degree	0.00639***	-0.00582***	-0.02075***
	(0.001)	(0.002)	(0.002)
Owner younger than 45	0.00784***	0.01588***	0.05275***
	(0.001)	(0.001)	(0.002)
Owner older than 65	-0.00335***	-0.01402***	-0.04734***
	(0.001)	(0.002)	(0.002)
Multiple owner	0.00028	-0.00758***	-0.00133
	(0.001)	(0.002)	(0.002)
Adopting any cloud computing	0.02239***	0.07995***	0.1346***
	(0.001)	(0.001)	(0.001)
Employee size: 10-250	0.00381***	0.02092***	0.04663***
	(0.001)	(0.001)	(0.002)
Employee size: > 250	-0.00534	0.04879***	0.04847***
	(0.006)	(0.008)	(0.009)
Manufacturing	0.02561***	0.08406***	0.02500***
	(0.002)	(0.002)	(0.003)
Family jointly owned	0.00644***	0.01809***	0.01550***
	(0.001)	(0.002)	(0.002)
In-house R&D	0.1515***	0.2476***	0.2421***
	(0.001)	(0.001)	(0.002)
Retirement-destination county	0.00092	0.0007	0.00498**
	(0.002)	(0.002)	(0.002)
Broadband adoption rate	0.00096	-0.00028	0.00744***
	(0.001)	(0.002)	(0.002)
Log(inventive class employment)	-0.00166***	-0.00143*	-0.00109
	(0.001)	(0.001)	(0.001)
Latent innovation index	0.0015	0.00745***	0.00275*
	(0.001)	(0.001)	(0.002)
Natural amenity scale	0.00040**	-0.00062**	0.00011
	(0.000)	(0.000)	(0.000)
Social capital index	-0.00001	-0.00421***	0.00351***
	(0.001)	(0.001)	(0.001)
Median home value ($1k)	-0.00001	0	-0.00005***
	(0.000)	(0.000)	(0.000)
Arts avocation index	0.01037***	-0.01783***	0.00421
	(0.003)	(0.005)	(0.005)

Active commute mode share	0.05921	-0.02959	0.2620***
	(0.046)	(0.061)	(0.069)
Population >25 with BA in Arts	0.01591	-0.00752	0.2140***
	(0.046)	(0.060)	(0.069)
Percent majoring science and engineering	0.02583*	0.04287**	0.01613
	(0.015)	(0.019)	(0.022)
Percent majoring STEM at 4-year-college	0.00012*	-0.00001	0.00026**
	(0.000)	(0.000)	(0.000)
Transit stops per 1k pop	-0.00005	0.00017	0.00018
	(0.000)	(0.000)	(0.000)
Eating places with 100 people w/ 2+ employees	1.48E-06	-0.00010*	-0.00013**
	(0.000)	(0.000)	(0.000)
SBIR award amount per 10K pop.	-4.64E-14	-8.38E-14	2.99E-13
	(0.000)	(0.000)	(0.000)

Notes: (1) The number of observations is 395000. (2) Significance levels: * 0.1, ** 0.05, *** 0.01. The ABS survey weights are used.